CENTRE FOR INNOVATION & SYSTEMS AI

Central TV Streaming Hub - User Operations Guide

User Manual: Web Portals

Target Audience: Operations & Presenters

1. Platform Overview & User Access Roles

The Central TV Streaming Hub provides a centralized system to manage, monitor, and stream content to all connected Smart TVs across corporate headquarters and remote branch offices. The system enforces Role-Based Access Control (RBAC) to separate fleet infrastructure management from day-to-day media operations and live broadcasting.

Portal Interfaces

- **Presenter Studio:** Real-time screen/camera broadcasting and direct media loop video uploads.
- **Admin Console:** Complete fleet discovery, TV settings, channel management, and emergency broadcast controls.
- **Executive Dashboard:** Real-time telemetry monitoring active TVs and streaming codecs.

Portal Access & Automated Routing

Authenticate at `http://<SERVER_IP>:8080/login.html` (or HTTPS `:8443`). The portal automatically evaluates your user role and directs you to the appropriate console.

1.1 User Roles and Permissions Matrix

User Role	Default Credentials	Permitted Capabilities	Restricted / Guarded Actions
Streamer / Operator Content & Broadcast	`streamer` `streamer123`	• Upload videos to continuous signage loop • Delete and reorder signage loop videos • Live screen sharing & camera broadcast (WebRTC) • Force TVs to switch between live stream & signage • Synchronize video playback across all screens	• Cannot delete or decommission TV displays • Cannot run network subnet discovery scans • Cannot create/modify channel definitions • Cannot trigger emergency alert banners
Administrator System & Fleet Control	`admin` `admin123`	• Full system access to all fleet management tools • Network TV discovery & auto-launch • TV display settings, reboots, and decommissioning • Emergency facility-wide alert dispatch • All streamer & media operations capabilities	*Unrestricted full administrative privileges across all APIs and endpoints.*

2. Using the Live Presenter & Media Studio

The Presenter Studio (`/presenter.html`) is the primary workspace for content operators and meeting presenters. It combines real-time low-latency video streaming with direct media loop management.

2.1 Live Broadcasting (Screen & Camera)

1. Log in as `streamer` (or `admin`) to enter the Presenter Studio.
2. Select your source: Click **"Share Laptop Screen"** or **"Camera & Mic"**. When prompted by your browser, grant media capture permissions.
3. Select the **Broadcast Destination**: Choose between *All Sites (HQ + Remote)*, *Headquarters Only*, or *Remote Site Only*.
4. Click the **"Start Live Broadcast"** button. Within 250 milliseconds, all target TVs will interrupt their current playback and switch seamlessly to your live video feed.
5. When finished, click **"Stop Broadcast"**. Target TVs will automatically return to the 24/7 digital signage loop without any manual intervention.

2.2 Signage Media Uploads & Playlist Loop Management

Streamers can manage the continuous video playlist directly from the Presenter Studio without requiring access to the administrative fleet console:

- **Upload Videos:** Choose one or more MP4, WebM, or MKV files using the file selector and click **"Upload Videos to Loop"**. Files are automatically verified and added to the round-robin loop.
- **Sync All to Video 1:** Click this button to simultaneously synchronize all TV displays across all offices to restart at the beginning of the playlist loop.
- **Force Signage Loop:** Immediately instructs any TVs currently on external channels to return to the corporate signage playlist.
- **Delete Videos:** Remove outdated announcement videos directly from the loop queue.

3. Using the Admin Console (Fleet Management)

The Admin Console (`/admin.html`) is restricted to the `admin` user and provides hardware-level fleet management.

Fleet Discovery & Telemetry

- **Auto-Scan Subnet:** Run `Scan & Auto-Connect Network TVs` to probe local subnets and automatically launch the TV player software on LG WebOS displays.
- **Remote Display Reload:** Send instant reload signals to refresh the player software on specific screens.
- **Prune Offline Displays:** Clean up decommissioned TV records with a single click.

Emergency Broadcast Override

- **Emergency Alert Banner:** Broadcast critical facility announcements with red audio-visual banners across every screen.
- **Channel Definitions:** Configure new streaming channels (RTSP cameras, dashboard web URLs, or custom WebRTC feeds).

4. Executive Dashboard & Telemetry

The Executive Dashboard (`/dashboard.html`) provides a high-level operational overview for IT NOC displays, showing connected display counts, current playback mode, and media server responsiveness.

5. Troubleshooting & Operations Reference

Symptom	Likely Cause	Resolution Procedure
Login failed or rejected	Incorrect credentials or locked account	Verify username and password (`streamer` / `streamer123` for media ops; `admin` / `admin123` for fleet). Wait 5 minutes if rate-limited.
Streamer redirected from Admin Console	Enforced RBAC security policy	Normal behavior. The `streamer` role is restricted to Presenter Studio. Use the `admin` account if fleet configuration is required.
Live broadcast shows black screen	WebRTC UDP packet block	Ensure internal firewall allows UDP port `8189` between presenter laptop and the central hub server.
Video loop skips or stutters	Unoptimized codec or high bitrate	Ensure uploaded videos use H.264 video codec at 1080p (1920x1080) 30fps. Unoptimized 4K high-bitrate files should be re-encoded.